Maria Jose Usma Rodas

Robotics Perception

3D Scene Understanding

Gaussian Splatting SLAM

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PROFESSIONAL SUMMARY

M.S. student in Electronic Engineering at Jeonbuk National University, working on robotics perception and 3D scene understanding. My research focuses on object-level semantic Gaussian mapping with SplaTAM, SAM2, and CLIP, connecting Gaussian Splatting SLAM, 3D reconstruction, and open-vocabulary scene understanding for robotics. Before graduate school, I worked in firmware and web development, which shaped my interest in building complete, reproducible systems

EDUCATION

Jeonbuk National University — Jeonju, South Korea
Master's Degree in Electronic Engineering

2024/09 - 2026/08

AI Robotics Laboratory, GKS Scholarship.

National University of Colombia — Colombia
Bachelor's Degree in Electronic Engineering

2016/02 - 2021/12

Graduated with the second-highest GPA in the cohort.

PROJECTS

Object-Level Semantic Gaussian Mapping with Temporally Consistent SAM2 Masks and SplaTAM — *Master's Thesis (Ongoing)*

- Developed a modular pipeline for object-level semantic mapping on Gaussian Splatting SLAM reconstructions using SplaTAM, SAM2, and CLIP.
- Propagated temporally consistent object masks across RGB-D frames with SAM2 and lifted masked depth pixels into 3D using camera poses and depth maps.
- Fused multi-frame object observations into object-level point clouds and applied voxel downsampling, statistical outlier removal, and DBSCAN clustering for noise reduction.
- Performed open-vocabulary semantic labeling with multi-view CLIP features extracted from selected object crops.
- Associated object identities and semantic labels with nearby 3D Gaussian primitives using KD-tree radius search, enabling

SKILLS

Programming: Python, C/C++, JavaScript

Models & Frameworks: SplaTAM, SAM2, CLIP

Core Techniques: RGB-D processing, point clouds, camera poses, 2D–3D

Tools: PyTorch, Open3D, Git, Linux, Docker, Qt, Pandas

AWARDS

GKS Scholarship 2023 - 2026

Awarded by the Korean Government.

LANGUAGES

Spanish: Native

English: Fluent

Korean: Intermediate (TOPIK II - level 3, 2024)

Research Interests

- Robotics perception
- Scene Understanding
- Gaussian Splatting SLAM
- Open-vocabulary semantic mapping.
- Object-level 3D reconstruction

CERTIFICATES

[DS4A / Colombia 6.0: Graduated with Honors, Correlation One 2022]

object-aware Gaussian map visualization without retraining the Gaussian map.
- Designed evaluation metrics for temporal mask consistency, 2D–3D geometric consistency, semantic confidence, and Gaussian-object association.

Data Science for All (DS4A) — *Ministry of Information and Communications Technologies of Colombia (MinTIC) and Correlation One (2022)*

- Selected as one of 1,800 participants from 15,000+ applicants for a national data science program.
- Worked with a team and government agency dataset to develop an algorithm for entity resolution and record matching under inconsistent and incomplete data.

EXPERIENCE

Kubo S.A.S, Colombia — *Web Developer*

2022/10 - 2023/07

- Developed and maintained web applications using frontend frameworks and REST APIs.
- Implemented features and supported maintenance tasks.

Titoma, Colombia— *Firmware Developer*

2020/11 - 2022/05

- Developed firmware in embedded C for ARM-based microcontrollers (STM32, ESP32, PIC).
- Implemented communication protocols (UART, SPI, I2C, Wi-Fi, Bluetooth).
- Worked with FreeRTOS, peripheral drivers, and real-time constraints.
- Built internal debugging and testing tools using Qt/C++ to support firmware development and hardware validation.
- Performed code reviews, testing, and technical documentation.

Titoma, Taiwan— *Embedded Systems Intern*

2020/02 - 2020/10

- Supported code development and version control using Git.
- Supported PCB design, component selection, and board debugging.